HACK APERTUS LIEBEFELD · BERN ECONOMIC DEVELOPMENT AGENCY CHALLENGE

BearMiner

Dig Bern gold from life-science partners.

A Research & Expertise Navigator that turns scattered university web pages into source-backed experts, units and partners — powered by Apertus.



THE PROBLEM

To find an expert today, you have to know **where** to look before you know **what** to look for.

"glucose sensors" | continuous glucose monitoring | CGM | biosensing | metabolic monitoring | wearable sensing

3

source ecosystems: UniBE subdomains, ARTORG, Diabetes Center Berne

5+

page types per institute: faculty, group, person, project, service

0

places a company can ask one question and get traceable answers

OUR SOLUTION

Two sides of the same search, meeting on evidence.

RESEARCH SIDE

What Bern can do

- Focused crawl of all three ecosystems
- Up to 500 ranked research highlights
- EN/DE medical keywords, verbatim evidence, source URL



MATCH

COMPANY SIDE

Who Bern should call

- Reads a pitch PDF, website or text, any language
- Builds a verified medical profile
- Collaboration potential 0–100 and P1–P4 priority

HOW IT WORKS

Public web in, source-backed answers out.

1

Collect

Inventory hosts only; robots.txt and rate limits respected

RULES

2

Select

Findings, projects, technologies, services, each with a quote

RULES

3

Understand

Need → terms and EN/DE synonyms; company medical profile

APERTUS

4

Retrieve & score

BM25 over records; five-part, recomputable match rate

RULES

5

Explain

Keeps relevant units and writes why each one matches

APERTUS

Apertus v1.5 — language understanding

Deterministic code — crawling, verification, scoring

WHAT A RESULT LOOKS LIKE

Every answer carries its own proof.

DIABETES CENTER BERNE · PROJECT

The Moonwalk Project

glucose · hypoglycemia

> “After the end of the study, algorithms will be developed to build a warning system for nocturnal hypoglycemia…”

dcberne.com/en/project/the-moonwalk-project/

Real record from our live crawl test: keywords, verbatim quote, public source.

SAMPLE COMPANY · WEARABLE CGM, ZURICH

81.1

Strong potential — prioritise outreach

Capability overlap	25.8/40
Partnership readiness	17.6/20
Domain fit	14.2/15
Swiss proximity	13.5/15
Signal depth	10/10

Why not higher: “Swiss company but no Canton of Bern site yet.”

PURPOSEFUL USE OF AI

Apertus where language is hard. Rules where trust matters.

Apertus does

Bridges vocabulary — “glucose sensors” becomes CGM, biosensing, metabolic monitoring, in English and German

Reads companies — any-language pitch to a structured medical profile

Explains matches — a short, sourced reason per result

Code does

Decides the score — same input, same number, fully offline

Checks every claim against the source text

Collects and ranks research without any model or GPU

TRACEABILITY BY DESIGN

No invented people, roles or projects.

VERBATIM GATE

Quote or drop

Each extracted item needs a quote found in the source (exact or ≥ 85% word match). Unsupported items are dropped and listed.

CLOSED CHOICE

No free-form people

The judge can only select candidate IDs it was given; each result keeps its source URLs.

HONEST GAPS

“No reliable match”

Below threshold, the system says so instead of guessing. Thin pages are marked insufficient.

PROVENANCE

Verified location only

Bern presence earns points only from a source, JSON-LD, imprint or Wikidata — never model memory.

SAFE INPUT

Injection defence

Web text is wrapped as untrusted data; sentences addressed to an AI are removed.

FAIR REACH

No size bias

No institutional quota, recency bonus or mention-count boost in ranking.

TECHNICAL RIGOUR

The score separates real fits from noise.

SAMPLE COMPANY	MATCH RATE
Bern glucose-sensor start-up, site in Burgdorf	88.6
Wearable CGM company, Zurich	86.8
Surgical navigation, Munich	76.1
Glucose diagnostics, Japanese-language source	52.1
Medical devices, little detail	38.2
One-line “wearable sensors” pitch	29.8
Fintech — out of scope	0.0

5 → 7 live medical pages produced seven evidence-backed records with keywords

30 labelled benchmark examples, 10 per source; build fails below 90% precision

EN·DE·JA sources handled — translated before matching

Tests for robots, redirects, budgets, evidence gates and the 500-record cap

SOVEREIGN · AFFORDABLE · EXTENSIBLE

Runs on Bern's terms, air-gapped if needed.

Sovereign

Apertus v1.5 on an on-premise or Swiss-cloud endpoint. SQLite and Python — no foreign closed service at runtime.

Low cost

Crawling and scoring need no GPU. LLM answers are cached, so re-runs are instant and identical.

Extensible

A new institution is one config entry — then re-run crawl and enrich. No pipeline rewrite.

```
# src/sources.json — add a source
{ "id": "newinst", "name": "New Institute", "url": "https://…", "institution": "…" }
```

FROM PROTOTYPE TO PILOT

Give us one quarter and one real company inbox.



WEEKS 1–4

Complete crawl

Full run over all three ecosystems on a Swiss-hosted Apertus endpoint.

WEEKS 5–8

Expert review

Agency staff review highlights; feedback tunes rules and synonyms.

WEEKS 9–12

Live outreach

Score incoming company requests and hand off P1 leads to Bern units.